

Finding Optimizations for Trading Using Search Economy Algorithms

Min-Yan Tsai , Jiang-Yi Zeng , Chia-Mu Yu , *Senior Member, IEEE*, and
Fan-Hsun Tseng , *Senior Member, IEEE*

Abstract—The development of artificial intelligence has led to the rapid growth of various industries. In addition to driving innovative applications, it has also brought significant changes to daily life. This transformation has accelerated the evolution of digital finance, enabling financial services to integrate AI and develop new applications. Among these services, smart investment has become one of the most popular. Due to the uncertainty of financial markets, achieving stable profits remains a challenging task. In this article, we propose a novel trading strategy that uses Search Economy Algorithms to analyze stock market trends. Our approach does not focus on a single stock; instead, it enables industry-wide investment decisions through our stock selection mechanism and fitness function, combined with the trading strategies provided by the Search Economy Algorithms. Compared with existing methodologies, our experimental environment is more complex. We benchmark our approach against ETF-0050 and demonstrate its effectiveness by evaluating capital growth rates and risk under specific parameters.

Index Terms—Assembly optimization, financial technology, metaheuristic algorithms, search economy algorithms.

I. INTRODUCTION

THE advancement of science and technology has led to a flourishing integration of information and daily life. In recent years, hardware computing power has been enhanced, and artificial intelligence has developed quickly. The numerous applications have led to significant breakthroughs, including financial ones. People focus on ways to generate profits in the financial market. Nowadays, there is much research that leverages artificial intelligence and historical data to develop methods that can achieve a return in the market. Examples of

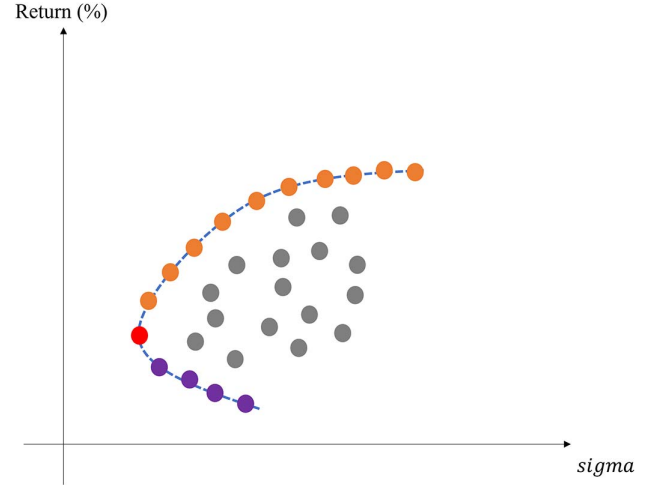


Fig. 1. Fintech service.

modern approaches include stock price prediction, identifying buy and sell points, and constructing the optimal stock portfolio. These methods differ significantly from traditional techniques. The traditional method considers risky financial products in the market. It creates a diversified investment portfolio through different weights. This method can generate multiple portfolios and efficient frontier curves [1], as shown in Fig. 1, the x-axis represents risk, and the y-axis represents a return. Various dots represent the risk and return of each asset in the portfolio at different weightings. The efficient frontier curve is represented by the blue line. In Fig. 1, the inefficient portfolio includes the purple and gray points. The efficient portfolio includes the red and orange points. However, it may not be possible to respond swiftly to rapid market changes; therefore, when extreme changes occur, this method may not be effective in delivering the desired benefits. As a result, this suggests that this methodology may be relatively insensitive to the market.

Previous research has proposed a factor model to obtain accurate stock market information. This concept can be used to estimate stock returns. The primary reason is that a stock's return has a multiplicative relationship with market movements. To predict stock market movements more accurately, researchers have developed the concept of factor modeling. Factor modeling is a methodology used to forecast the expected returns

Received 30 December 2024; revised 5 March 2025; accepted 14 April 2025. Date of publication 3 June 2025; date of current version 3 December 2025. This work was supported in part by the National Science and Technology Council (NSTC) in Taiwan under Contract NSTC113-2221-E-006-221-MY3, NSTC111-2222-E-006-012-MY3. (Corresponding author: Fan-Hsun Tseng.)

Min-Yan Tsai is with the Department of Information Management and Finance, National Yang Ming Chiao Tung University, Hsinchu 30010, Taiwan (e-mail: wer9u623@gmail.com).

Jiang-Yi Zeng and Fan-Hsun Tseng are with the Department of Computer Science and Information Engineering, National Cheng Kung University, Tainan 70101, Taiwan (e-mail: p78121069@gs.ncku.edu.tw; tsengfh@gs.ncku.edu.tw).

Chia-Mu Yu is with the Department of Information Management and Finance, Department of Electrical Engineering, National Yang Ming Chiao Tung University, Hsinchu 30010, Taiwan (e-mail: chiamuyu@nycu.edu.tw).

Digital Object Identifier 10.1109/TCSS.2025.3563455

of marketable securities. Its fundamental assumption is that a linear relationship exists between stock returns and market movements. Through regression analysis, the value of the intercept term, denoted as α , can be examined to assess whether a stock is overvalued or undervalued. Based on this, multi-factor models have been developed, the most well-known of which is the three-factor model. The three-factor model considers market risk, company size, and the book-to-market ratio to predict stock returns. The renowned three-factor model [2] is constructed by considering company size, market capitalization ratio, and prevailing market conditions. However, the factors in the factor model carry significant economic meaning, and each factor may represent a different signal, making it difficult for nonprofessional investors to understand and effectively utilize them. Among the emerging methods, quantitative trading is becoming more widely used. This trading technology is often based on mathematical models and data analysis techniques to search for possible investment strategies. Then, it uses back-testing to evaluate these investment strategies. The development of AI allows many researchers to conduct financial research. For instance, the research compared the predictive performance of long short-term memory (LSTM) and the autoregressive moving average model (ARMA) prediction effect. The results indicate that a more precise prediction of prices can be achieved using LSTM [3]. Several studies have also explored hybrid models forecasting stock prices. Shah et al. proposed a hybrid model combining BiCuDNNLSTM and 1-D-CNN that was developed to predict stock prices [4]. Wu et al. adopted the opening price, closing price, and other relevant stock information, which were combined into a single image to predict stock prices using convolutional neural networks (CNN) and LSTM models. This hybrid modeling approach is expected to improve prediction accuracy [5]. There have also been studies indicating that social media articles and messages have a fluctuating impact on cryptocurrency prices. Consequently, social media data has been used to analyze and predict these price movements [6]. According to the aforementioned research, a strategy can be designed more effectively if the price is known in advance. This strategy can help achieve higher returns. However, the market has many noises. The AI model cannot catch every noise, making it difficult to predict the stock price accurately. The small price difference can result in a significant loss to a portfolio. Therefore, this suggests that there is still a risk of inaccurate stock price prediction. To effectively use AI for productive investment, research uses quantitative trading as a starting point and artificial intelligence models to build a set of trading strategies and methods. In [7], a method based on the pairs strategy is proposed. The major idea is to use a threshold as the buy-and-sell signal. This approach consists of two stages, both of which use deep learning. In the first stage, the optimal threshold is selected, and in the second stage, nonreturning trades are identified and deleted based on the first set of thresholds. In addition, there are also studies that expect the machine to be able to design effective investment strategies just like a financial expert through reinforcement learning. In recent years, the transformer model [8] has become very popular because GPT-3 [9] uses a transformer decoder as a major model.

As a result, extensive research has been conducted using the Transformer model. The BERT model has been used to analyze the sentiment of news headlines, followed by the use of a GAN to predict stock prices [10]. The BERT model has also been applied to analyze investor sentiment before the market opens. Aggregated stock quotation data and sentiment data are used to predict the next trading day's closing price via an LSTM model [11]. Furthermore, ChatGPT has been explored as a tool for stock price forecasting, with results indicating that its predictions outperform traditional sentiment analysis in terms of stock returns [12]. Furthermore, some studies propose two training mechanisms to achieve their objectives. The issue that most strategies can only be applied in a single market has been addressed [13]. With the rapid advancement of technology, many novel approaches have contributed to the progress of FinTech. However, they also demand greater computing resources and corresponding support. For instance, artificial intelligence models necessitate substantial amounts of data and incur high computational costs. Although large language models (LLMs) have significantly improved many applications, their computational resource and data requirements are much greater than those of previous approaches [14]. Even CNN or RNN models often require large amounts of data and a high cost of computing. The neural network model also may not be the most suitable option due to its operating environment, which is characterized by limited data and constraints on computational power. Although traditional methods are less energy-intensive, their performance may not be as effective as that of these newer methods. To strike a balance between resources and effectiveness, this article focuses on a smart investment approach that identifies effective investment strategies using limited stock price data. In this approach, a meta-heuristic algorithm [15] is employed to solve the problem. Many contemporary research methods also leverage this information to construct investment portfolios; however, existing approaches may lack flexibility. Given the dynamic nature of financial markets, constructing an effective strategy portfolio may be challenging. Therefore, this article proposes a useful strategy portfolio utilizing technical information and assesses its effectiveness through various methods. We use SE's algorithm for stock selection and trading strategies within one of Taiwan's industries. We propose a mechanism to first create a portfolio pool of stocks with the potential for appreciation and then to buy and sell based on the performance of the individual stocks within that portfolio. Here, a trading strategy is developed using historical data to evaluate its performance against future data. In the end, we expect that the investor's objective is to beat the market. We chose the Taiwan ETF-0050 for comparison, and we compared the capital growth rate to two indicators of risk. The final results indicate that our mechanism is effective.

This article is organized as follows: in Section I, we describe the existing overview of smart investing and briefly describe our methodology; in Section II, we explain the background knowledge of Metaheuristic algorithms and Search Economy Algorithms, we also delve into the related research on Metaheuristic algorithms for the stock market. In Section III, we describe our methodology; in Section IV, we describe the experimental

TABLE I
SUMMARY OF STUDIES INVOLVING FITNESS EVALUATION, DATASETS, AND ALGORITHMS

No	Fitness	Dataset	Test Time	Algorithm	Sliding Windows
[20]	Trend Ratio	ETF-0050	2010-2016	GA, GQTS, NQTS, ANGQTS	Yes
[21]	Trend Ratio, Emotion Index	DJ30	2010-2019	GQTS, ANGQTS	Yes

setup; in Section V, we present the results; and in Section VI, we present the results of the study. Section VI summarizes the article.

II. BACKGROUND AND RELATED WORK

A. Search Economy Algorithms

Search economic (SE) algorithms [16], [17] are a type of metaheuristic algorithm, and play a significant role in the search for solutions within the field of artificial intelligence. The primary feature is the ability to generate a combination of optimized solutions in any field within a certain period of time. Metaheuristic algorithms have many branches. Usually, they are grouped according to the number of candidates and the number of solutions. As the name suggests, if there is only one candidate, it is considered a single solution. The multi-solution method involves having more than one candidate. It can be further categorized into Swarm intelligence optimization algorithms (SIOP) [18] and non-SIOP design methods. The difference lies in whether the method is designed for a particular group, e.g., ants, bees, etc. The famous particle swarm optimization (PSO) [19] belongs to multiintelligence, and differential evolution belongs to nonmultiintelligence.

The SE concept divides the solution space into n regions. Each region has a certain number of candidate solutions and can be seen as a different market product portfolio. Candidate solutions are for stock market portfolios. The method will be updated through searchers. This engine can be seen as an investor. The engine also has a market product portfolio. The method will generate a new portfolio based on the portfolio on hand by exchanging information with portfolios in the market. The updating methods can be referenced to genetic algorithms for crossover and mutation methods for updating. The searcher will go to different regions and exchange information with the solutions selected in each region, and the engine will choose the best solution based on all the available solutions. On the other hand, the method records which region of solutions was selected by the searcher. This prevents the engine from spending too many resources on searching through a region. Because searching too long in one region may cause the final selected solution to fall into a local minimum. Thus, this method allows for a deeper search of the solution space than other methods, so that more solutions can be searched to find more combinations.

B. Algorithmic and Stock Market Related Research

Stock market portfolio optimization, which extensively utilizes metaheuristic algorithms. In this section, we will explore

research on the application of metaheuristic algorithms in stock investment and portfolio optimization. Chou et al. proposed a trading ratio to create a portfolio. This method uses ANGQTS as an optimization algorithm. To Find the trend ratio for each stock by using past stock prices. This method tries to combine the combinations with the highest trend ratios, which usually yield relatively good returns [20]. Building on this foundation, ANGQTS was further applied as an optimization algorithm in another study, which utilized the trend ratio and sentiment indices as optimization criteria to construct a portfolio [21]. A detailed comparison of the portfolios based on the above ratios is shown in Table I. Another common method used in the metaheuristic Algorithm is to use technical indicators to find the corresponding buy and sell time points. Kuo et al. use QTS to optimize the MA technical indicators to find the best entry and exit points for a stock. This approach aims to maximize returns over a given period. The strategy was subsequently applied in future trading. Additionally, the study evaluated the effectiveness of WMA and Eva [22]. Moreover, the global best-guided quantum-inspired tabu search algorithm (GNQTS) was employed for optimization. The objective was to determine the threshold values for overbought and oversold conditions. Here, the values are obtained through the RSI with the intention of optimizing three parameters, namely overbought, oversold, and the RSI parameters [23]. Another approach concentrated on optimizing two key technical indicators: MACD and RSI. The optimization objective is to maximize returns, minimize transaction costs, and reduce risk, including the VIX index, while also considering the overall performance of all objectives. The algorithm is optimized via NSGA-II [9]. In addition, optimization methods such as dynamic vector-evaluated particle swarm optimization (DVEPSO), multiobjective particle swarm optimization with Crowded Distance (MOPSO-CD), and dynamic nondominated sorting genetic algorithm (DNSGA-II) are used to optimize the parameters of RSI and MACD indicators. The dataset is divided into training datasets and testing datasets and validated by a sliding window [24].

Another study introduces the concept of multistrategy optimization. The author uses GA as the optimization algorithm. The gene sequence comprises multiple trading strategies, portfolios, and weight allocations. The trading strategy portfolio comprises multiple trading strategies. The optimization objective includes the trading strategy portfolio, weight allocations, and maximum return. The primary considerations in this context are return, risk, and solution diversity. Simultaneously, solution diversity is incorporated into the fitness function. This approach aims to mitigate asset bias in specific portfolio allocations [25]. An alternative approach to optimizing trading

TABLE II
SUMMARY OF OPTIMIZATION APPROACHES AND DATASETS USED IN VARIOUS STUDIES

No	Fitness (f)	Technical Indicators	Dataset	Test Time	Optimization Technical Indicators	Algorithm	Sliding Windows
[22]	$f = \text{return}$	SMA, WMA, EMA, MA	(TW) ETF-0050, 2330, 2317, 2412, 2201, 2409, 2105, DJIA, IXIC	2014/1-2017/12	Single Objectives	GQTS	Yes
[23]	$f = \text{return} \times \text{cost}$	MACD, RSI	USDZAR currency pair	2011-2017, 2015-2019	Single Objectives	DVEPSO, MOPSO-CD, DNSGA-II	Yes
[9]	$f = \text{profits} \times \text{cost} \times R_t \times R_{vix}$	MACD, RSI	VIX, S&P500	2000/1/1-2010/12/31	Single Objectives	NSGA-II, SPEA2	No
[24]	$f = \text{return}$	RSI	DJIA, INX, NASDAQ, NYSE, AAPL, CSCO, IBM, INTC, MSFT	2010/1-2020/12	Multiple Objectives	GNQTS	Yes
[25]	$f = \text{return} \times \text{risk} \times \text{weight (balance)}$	MA, MOM, KD, CCI, PSY, DIF, DI, MACD	Sideways trend dataset	2011/1/1-2016/12/31	Multiple Objectives	GA	Yes
[26]	$f = \text{return}$	Technical indicators pool	S&P500, DJIA, IXIC, NYSE	2005/1/3-2010/4/21, 2010/5/3-2013/2/28	Multiple Objectives	QTS	Yes
[27]	$f = \text{profits} \times \text{transaction cost} \times \text{risk}$	EMA, RSI, MACD, Larry Williams %R	IIBOVESPA index	2005-2014	Single Objectives	SPMODE	No
[28]	$f = \text{annual return} \times \text{Sharpe ratio}$	DEMAC, RSI, MACD, MARSI	DJIA	1982/1/1-2012/1/1	Multiple Objectives	GA	Yes

strategies entails predefining a large set of trading signals. The signals are generated based on technical indicators, and multiple signals collectively form a strategic trading pool in which trading signals are paired. The main purpose is to test which technical indicators in the pool can generate effective trading signals within a specific timeframe, and a sliding window is also used to verify their validity [26]. Additionally, previous studies have also examined the effectiveness of multiple indicators individually. In the design of the fitness function, not only returns and risk but also the number of trades are taken into account, achieving multiobjective optimization [27]. Finally, another study focused on optimizing multiple indicators by selecting four default indicators. This approach optimizes the duration for each indicator, determines the optimal number of days for each, and constructs a trading strategy [28].

Our objective is to assist investors in achieving substantial capital growth in the stock market. To achieve this, we concentrate on stocks within the semiconductor industry in the Taiwanese market. We create stock selection mechanisms and multiple technical indicators to construct industry-specific investment portfolios and develop corresponding trading strategies. In contrast to other studies, most focus on a single stock or an index to build a trading strategy, and many focus on using an index as a reference for trading signals. However, in today's rapidly evolving financial market, relying on a single indicator over an extended period may prove ineffective. Therefore, our methodology uses multiple technical indicators to identify optimal market entry strategies. In contrast to existing methodologies, we focus on stock selection and trading through the use of multiple indicators.

III. METHODOLOGY

Before searching for a strategy, we will first select stocks within the industry and determine whether to include them in our portfolio based on their past performance. Initially, we must

identify trending stocks, assuming that a positive trend in the past may indicate a likelihood of continued growth in the future. Based on this premise, we define trending stocks, which are calculated as follows:

$$e_i = \sum_{\text{day}=2}^n \text{close}_{i,\text{day}} - \text{close}_{i,\text{day}-1}. \quad (1)$$

We define the trend stock value as e . The variable e_i denotes the expected value of the i th trend stock. If the expected value of a stock is greater than 0, the stock is seen as a trending stock. The value is used as a closing price for calculations, which take the closing price of the previous day, subtract the closing price of today, and then accumulate the results. Ultimately, we will invest in these trend stocks using the SE to calculate the most appropriate trading strategy for investing in these stocks.

We propose a method that uses multiple indicators to generate buy-and-sell signals. The indicators are the moving average (MA), exponential moving average (EMA), relative strength index (RSI), and bias ratio (BIAS). Thus, we collect all values of each indicator for every day. For example, when $\text{MA5} > \text{MA20}$, it indicates that the stock has strong short-term bullish momentum. Therefore, we use it as a signal to buy the stock. When $\text{MA5} < \text{MA20}$, it may indicate that the upward trend is coming to an end. Therefore, we use it as a signal to sell the stock. In this way, we aim to identify the optimal entry and exit points for each of the four technical indicators over a specific period. In this method, MA is often used as one of the key indicators that most people refer to. This is calculated by averaging the prices from the previous days.

Therefore, this indicator reflects the recent buying sentiment of the stock. If the MA line trends upward, it may indicate the market has confidence in the stock. The EMA is more price-sensitive than the MA. Since the EMA assigns different weights to different days, errors may lead to even larger discrepancies. The RSI is calculated using the average gain and average loss

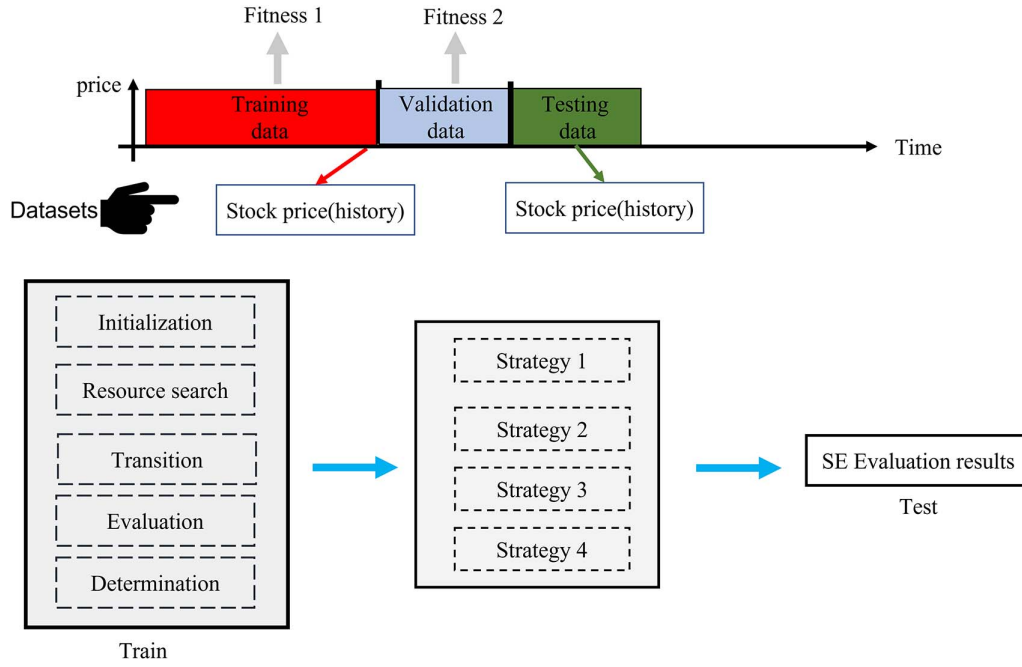


Fig. 2. SE transaction calculation framework.

over the previous n days. Then, the average gain is divided by the sum of the average gain and average loss. The average increase can be used to reflect whether the market is actively trading a stock. Usually, 20 and 80 are used as benchmarks. Above 80 means too hot, and below 20 means it is relatively quiet. However, the RSI may be passivized.

The usual rule is to sell over 80 and buy 20, but it is also possible that the stock price continues to rise when the numerical value is at 80, and if an investor chooses to take profits at this time, he or she will lose the opportunity for the stock price to continue to rise, which is also known as passivation. The BIAS indicator is used to monitor the fluctuations in price increases and decreases, as well as the deviations generated by the MA. This indicator can indicate whether the stock is overbought or oversold. In summary, indicators have certain limitations, and it is essential to recognize that we cannot depend on a single indicator as a trading signal. To achieve a substantial increase in the rate of capital growth, this method uses the four aforementioned indicators to analyze trading signals. Therefore, this method uses the four aforementioned indicators to generate trading signals, aiming to identify the most effective indicators for trading across different time periods. In this article, we use the SE as our optimization method, which possesses robust global search capabilities. Our objective is to extract strategies from historical data that can guide future investment decisions. Our framework can follow Fig. 2. First, we divide the dataset into training and testing and specify a fixed length for the training data. Subsequently, the most effective strategy is identified using SE. Our objective is to maximize returns by formulating trading strategies derived from historical data, thereby achieving optimal capital growth. To mitigate the risk of a strategy performing well in the past but failing in the future, we define two fitness functions.

A detailed explanation of the calculation will be provided in the following paragraph, where the most effective trading strategy will be identified based on fitness values. As the method uses four technical indicators as trading signals, the optimal strategy can incorporate up to four indicators.

To obtain the maximum, we design a fitness, and it can follow.

Subject:

$$\begin{aligned} & \text{maximize } F \\ & F = w * \text{Fitness}_1 + (1 - w) * \text{Fitness}_2 \\ & \text{Fitness}_b = \sum_{i=1}^s \text{Return}_i, \quad b \in \{1, 2\} \end{aligned}$$

where

$$\text{Return}_i = \sum_{t=1}^4 \frac{\text{sell}_{i,t2,j} - \text{buy}_{i,t1,j}}{\text{buy}_{i,t1,j}}$$

and

$$\begin{aligned} & t1_{t,1} < t2_{t,1} \\ & t_i = \{(t1_{t,1}, t2_{t,1}), \dots, (t1_{t,j}, t2_{t,j})\} \end{aligned}$$

As previously mentioned, if a trading strategy only produces favorable results for a limited period, it may suggest that the strategy has converged to a local optimum. Our goal is to maximize the fitness function F while addressing the potential challenges outlined above. Therefore, we design two fitness functions to compute fitness values, denoted as Fitness_1 and Fitness_2 . Given a batch of data, we divide it into three equal parts: training data, validation data, and testing data. SE focuses on the training dataset to identify the optimal investment strategy and uses the validation dataset to assess both successful

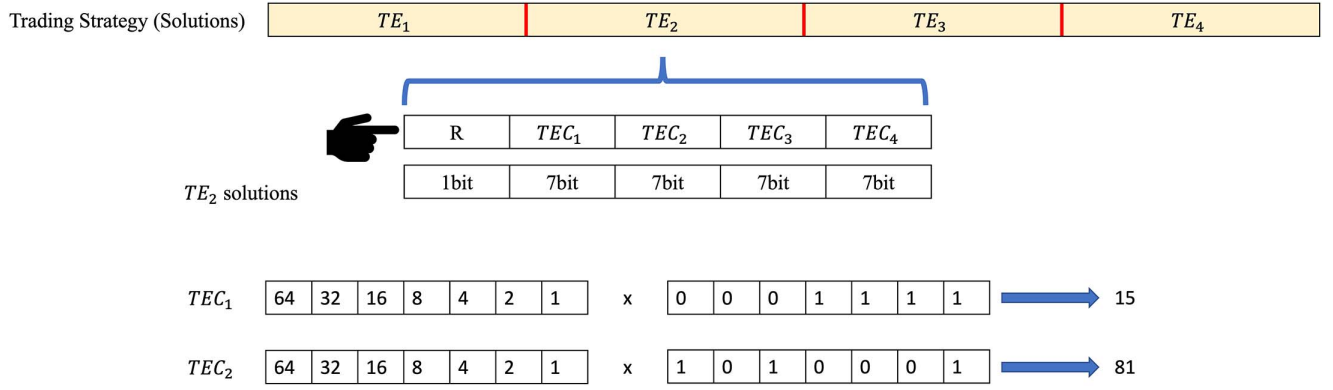


Fig. 3. SE encoding rules.

and unsuccessful outcomes. $Fitness_1$ represents the investment profitability of the training dataset, while $Fitness_2$ corresponds to the investment profitability of the validation dataset. We design a weighting factor, w , to determine which time period's results should be assigned greater importance, allowing for a flexible design. We evaluate our trading strategy using this fitness function and apply the optimal results to assess our test data. We consider each fitness function to be an important parameter in the calculation process. For the sake of clarity, we will set w to 0.5 to illustrate the calculation process.

The returns of each stock s are summed to obtain $Fitness_b$. The value of $Fitness_b$ can correspond to either $Fitness_1$ or $Fitness_2$, as both are computed using the same method, differing only in whether they are derived from the training dataset or the validation dataset.

The return of the i th stock in a single trade is defined as $Return_i$. The return is determined by the buy and sell prices, which vary based on the specific stock and the trading signals generated by different trading strategies. To systematically represent these trades, we provide a detailed introduction. The variable $buy_{i,t1,t,j}$ denotes the buy price of stock i on day $t1_{t,j}$ and the variable $sell_{i,t2,t,j}$ denotes the sell price of stock i on day $t2_{t,j}$. The variable i represents the stock index, indicating different stocks in the dataset. The variable t denotes the technical indicator, as we use four distinct trading strategies. The variable j indicates the transaction number, sequentially numbering each executed trade for ease of reference.

To calculate returns, we pair each buy signal $t1_{t,j}$ with a corresponding sell signal $t2_{t,j}$. Suppose the training period spans from day 1 to day 40. The identified buy signals appear on days $\{20, 25, 31, 37\}$, while the corresponding sell signals on days $\{21, 22, 27\}$. The first buy signal is observed on the 20th trading day from the beginning of the training period. Since a trade cannot be executed immediately upon signal generation, it is carried out on the following day. That is $t1_{1,1} = 21$, which is matched with the closest sell signal occurring on day $t2_{1,1} = 22$. This results in the trade pair $(t1_{1,1}, t2_{1,1}) = (21, 22)$. The second buy signal is observed on the 25th trading day, i.e., $t1_{1,2} = 26$, and is matched with the corresponding sell signal on day $t2_{1,2} = 28$, resulting in the trade $(t1_{1,2}, t2_{1,2}) = (26, 28)$. The buy signals on days 31 and 37 do not have corresponding

sell signals within the available period. In such cases, the sale is executed on the final day of the training period (day 40). As a result, these trades are recorded as follows:

$$t_i = \begin{cases} (t_{1,1}, t_{2,1}) = (20, 21), \\ (t_{1,2}, t_{2,2}) = (25, 27), \\ (t_{1,3}, t_{2,3}) = (32, 40), \\ (t_{1,4}, t_{2,4}) = (38, 40). \end{cases}$$

The trading strategy for the test data will be determined by identifying the maximum fitness value. The concept of SE is analogous to the financial market, which comprises both investors and financial instruments. Investors seek to identify the optimal investment portfolio. Because the market changes rapidly, investors must adjust their strategies according to the prevailing market conditions to avoid excessive losses. In the SE framework, the solution space is partitioned into distinct regions. This can be regarded as a combination of various financial instruments, including the stock market, bond market, and futures market. Investors can select the markets in which they wish to invest. SE defines the searchers ($Searcher_i$) and the candidate solutions $good_k$. These two elements represent investors and financial instruments in the market, respectively. The combination of solutions is enhanced by optimizing both the searcher and the candidate solutions. In this article, we employ binary encoding to represent the solution. The solution comprises four indicators, denoted as TE_1 TE_2 TE_3 TE_4 . Each signal consists of an R parameter and four day-related parameters as in Fig. 3. R accounts for the possibility that each technical indicator may fail under certain conditions. Consequently, each technical indicator is defined as a strategy.

When R equals 1, the gains of the sensory-technical indicators are incorporated into the fitness value calculation. Fig. 3 illustrates the trading strategy based on four technical indicators. Each technical indicator trading strategy comprises two substrategies: one for buy signals (TEC_1 and TEC_2) and one for sell signals (TEC_3 and TEC_4). Assuming that TEC_1 is 0001111 and that TEC_2 is 1010001, their decimal equivalents are 15 and 81, respectively. In the context of the EMA trading strategy, TE_2 generates a buy signal when $EMA_{15} > EMA_{81}$. The sell signal follows the same principle. In this case, we use 7 bits for representation; however, the exact number of

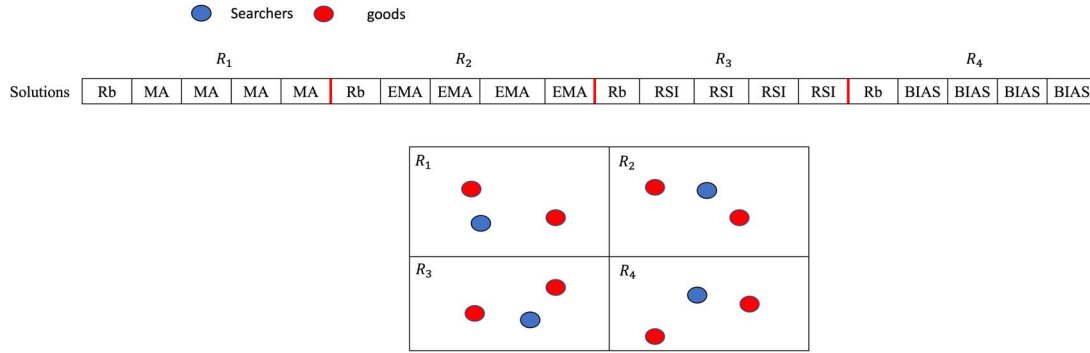


Fig. 4. SE resource allocation.

bits is determined by the number of days in the testing data. If the number of test days is 60, the number of bits is 6; if the number of test days is 120, the number of bits is 7; and if the number of test days is 240, the number of bits is 8. The reason for this discrepancy is that when the number of indicator days exceeds the number of test days, these indicators may have limited effectiveness. Therefore, we exclude values that are excessively divergent. For instance, when the number of test days is 60, the trading strategy does not satisfy the condition $MA_{63} > MA_{80}$. In our system, we use SE to identify the most effective investment signals. We segment the search space into four distinct regions. Each region represents one of the following strategies: MA, EMA, BIAS, and RSI. The regions are denoted as R_1 , R_2 , R_3 , and R_4 , as shown in Fig. 4. We allocate searchers to each region implementing the randomized approach. Consequently, the number of searchers may vary across regions. The number of candidate solutions is set to k in each subspace. The search engine will then traverse each region and update each candidate solution to generate a new one. During this generation process, crossover and mutation are restricted to specific regions. For example, as shown in Fig. 5, searcher i in the R_1 region is exchanged with $good_1$ elements. In this case, exchange and mutation are restricted to the region marked in orange. The R_1 region governs the MA indicator; similarly, R_2 is associated with the EMA, R_3 corresponds to the RSI, and R_4 represents the BIAS. The crossover uses a single-point exchange, as shown in Fig. 6. Suppose that searcher i wants to generate a new solution with the k th good element in region j . This method randomly selects a point as the exchange point. Fig. 6, the section of the solution to the right of the red dotted line is swapped between two solutions, while the left side remains unchanged. The mutation is performed by randomly selecting a position within the solutions, as shown in Fig. 6, and the updated sections are highlighted in blue font.

Each searcher selects the optimal combination based on the new solution. If the best solution outperforms the one currently held by the searcher, it replaces the existing solution; otherwise, the original solution is retained, and the process repeats in the next round. After updating the searcher, the available candidate solutions in the market include both new and existing options. In this step, the top α solutions are first selected from all candidates. Then, β solutions are randomly chosen from these

top α to advance to the next generation. Finally, the system tracks the regions each searcher has visited in this generation to prevent repeated searches in the same region.

The preceding description outlines the overall mechanism. This subsection explains how to select searchers to distinguish between effective and ineffective solutions. Since the mechanism divides the solution space into four regions, searchers aim to move to the region with the highest expected value to select a new solution, as shown in the following equation:

$$E_{ij} = f_1(M_j)f_2(v_j^b)f_3(\rho_j). \quad (2)$$

Each searcher has specific expectations for the values in each region. Thus, E_{ij} denotes the expected value of the i -th searcher in the j th region. The expected values consist of three components: search heat, search efficiency, and historical performance. The first component, search heat, can be obtained using (3). The variables t_j^a and t_j^b are positive integers, where t_j^a denotes the number of consecutive searches in region j

$$f_1(M_j) = \frac{t_j^b}{t_j^a}. \quad (3)$$

For example, if region Two is searched by three different searchers, and they enter the region to select a new solution, the value of t_2^a will increase by three, while t_2^b remains unchanged. If no searcher selects region 1 in the current generation, then t_1^a is set to 1, and t_1^b is incremented by one. The purpose of this is to prevent searchers from converging on the same region for multiple generations, which may hinder their ability to find the best solution in the entire domain. Search effectiveness refers to how well each searcher generates new solutions in each region

$$f_2(v_j^i) = \frac{\sum_{k=1}^{\omega} f(v_{j,k}^i)}{\omega}. \quad (4)$$

If the searchers find better solutions, then the likelihood of discovering even better solutions in the future increases, as shown in (4). The variable v_{ji} denotes the performance of the i th searcher in the j th region. The calculation method involves summing the fitness values of all new solutions generated by the searchers and then dividing the total by the number of new solutions. Finally, historical performance evaluation assesses



Fig. 5. SE element exchange limit.

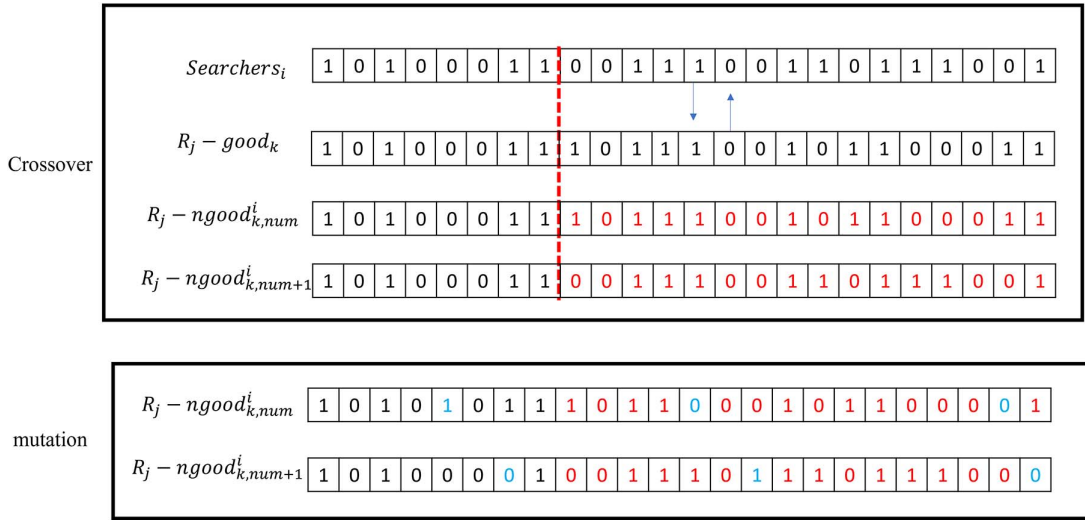


Fig. 6. SE crossover and mutation schematic.

past performance in the region. Since the region has demonstrated strong performance in the past, it is likely to present favorable opportunities in the future. If the selection evaluation relies solely on the new solution without considering historical performance, there is a higher risk of converging to a local optimum. However, incorporating historical performance can effectively mitigate this issue

$$f_3(\rho_j) = \frac{f(m_j^g)}{\sum_{j=1}^h f(v_j)}. \quad (5)$$

Therefore, the formula for the final expectation evaluates the historical performance of each region in generating solutions. The historical performance can be computed using the formula shown in (5). The variable m_j^g denotes the best performance of the solution in the j th region in the past. The variable v_j denotes the sum of the fitness values of all candidate solutions in the j th region.

Overall, SE prevents convergence to a regional optimum during the search process by calculating the expected value. This approach facilitates the discovery of a more diverse set of

solutions compared with other existing methods, which is why we chose SE for searching trading strategies.

IV. EXPERIMENTAL SETTING

We obtained the stock dataset from the Taiwan Economic Journal (TEJ) and selected the Taiwanese semiconductor industry as our experimental dataset. We selected 48 stocks for this experiment, and the dataset spans the period from 2012 to 2023. A sliding window approach is employed to validate the test periods, with the window length determined by the duration of the test. We used training datasets of 240 and 120 days for the test. The testing dataset consists of periods of 60, 120, and 240 days. The test duration is shorter than the training period. The dataset is divided into training and testing data. This study uses training data to identify signals conducive to initiating a trading strategy and employs test data to evaluate the effectiveness of signals detected by SE in identifying stable and highly profitable trading strategies for future implementation. The sliding window lengths are 240, 120, and 60 days, respectively.

Here, we classify the three mechanisms into three distinct categories: standard, risk-focused, and market trend-focused.

We refer to these as standard mode (ST), market mode (market), and risk mode (risk). “ST” means that if a trend stock exists, it will be traded during the test period. “Market” means that if a trending stock exists and the mechanism determines that the trend is positive, it will hold the stock at the time of purchase until the last day of the test period. “Risk” takes into account the risk of selling a stock if the holding loses more than expected. We tested thresholds of 5%, 10%, 20%.

In the previous paragraph, we discussed trend stock parameters and methods for calculating fitness. In our experimental results, a trend stock is defined as having a parameter e that is greater than or equal to 0. As mentioned earlier, our fitness function consists of two subfitness components. Here, we assign equal weights to these components, setting w to 0.5. Our goal is to achieve a higher capital growth rate using our strategy compared with ETF-0050 in all scenarios. The mechanism functions as an active investor with a positive impact.

Therefore, we define a 20-day buy-and-hold interval beginning on any day within the first month of the test period. Next, we calculate the capital growth rate, which represents the cumulative 10-year return on investments made during this period. Finally, we present the average capital growth rate within this framework for comparative analysis. To compare performance, we use the Capital Growth ratio to evaluate the effectiveness of the strategies. We use SE to compute the strategy performance using (6). AS our method updates the strategy periodically, multiple rates of return will be generated during the test, depending on the frequency of strategy changes. We then aggregate these values to compute the overall fund growth rate. Because 0050 is a long-term index ETF, we use the same timing for buying and selling in the last month of the test period. Additionally, we apply the dollar-cost stocking (DCS) method, and the final fund growth rate is calculated using (7). Since the SE strategy closes at a fixed time, we define risk as the lowest profitability point in the entire backtesting process. In contrast, ETF-0050 uses the maximum drawdown (MDD) as the risk benchmark. MDD is determined by identifying the maximum cumulative return (peak) during the entire test period, denoted as H , followed by the subsequent lowest value, denoted as L . The formula for calculating MDD is given by the following equation:

$$\text{CapitalGrowth}_{SE} = \sum_{t=1}^y R_t \quad (6)$$

$$\text{CapitalGrowth}_{DCS} = \frac{\text{SellAmount} - \text{Buycost}}{\text{Buycost}} \quad (7)$$

$$\text{MDD} = \frac{H - L}{H}. \quad (8)$$

V. EXPERIMENTAL RESULTS

In the experimental results, we compare training periods of 120 and 240 days to evaluate the trading strategy’s performance under different test durations.

As shown in Tables III and IV, the results indicate that when the number of training days is set to 120, the final capital growth rate is negative regardless of whether the number of testing days is 60 or 120. We deduce that, since the data used is daily, the training period is too short for SE to effectively capture trading

TABLE III
TRAINING DATA IS BASED ON THE RESULTS OF
120 DAYS OF BACKTESTING, AND THE
FOLLOWING VALUES ARE SHOWN IN %

ST	60	120
Average capital growth	−15.49	9.4
Best capital growth	−14.73	13.9
Worst capital growth	−18.08	7.41
Market	60	120
Average capital growth	−16.34	14.94
Best capital growth	−14.73	17.37
Worst capital growth	−18.08	11.3
Risk5	60	120
Average capital growth	−16.254	16.08
Best Capital Growth	−15.49	16.94
Worst capital growth	−18.04	14.82
Risk10	60	120
Average capital growth	−16.64	15.02
Best capital growth	−15.68	19.42
Worst capital growth	−17.62	10.86
Risk20	60	120
Average capital growth	−17.13	15.5
Best capital growth	−16.3	18.2
Worst capital growth	−17.7	13.63

TABLE IV
TRAINING DATA IS BASED ON THE RESULTS OF 240
DAYS OF BACKTESTING, AND THE FOLLOWING
VALUES ARE SHOWN IN %

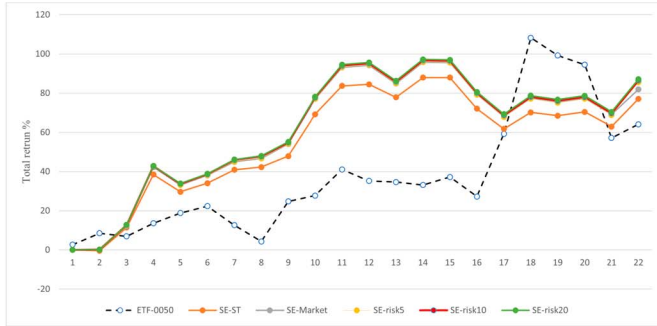
ST	60	120	240
Average capital growth	−16.34	76.17	15.44
Best capital growth	−14.73	77.11	16.57
Worst capital growth	−18.08	75.17	13.25
Market	60	120	240
Average capital growth	−16.34	85.46	14.86
Best capital growth	−14.73	86.93	22.03
Worst capital growth	−18.08	81.92	11.12
Risk5	60	120	240
Average capital growth	−10.27	86.25	14.89
Best capital growth	−9.83	87	20.08
Worst capital growth	−10.57	85.53	11.97
Risk10	60	120	240
Average capital growth	−10.18	85.96	14.76
Best capital growth	−9.6	87.05	16.48
Worst capital growth	−10.58	85.31	12.42
Risk20	60	120	240
Average capital growth	−9.96	86.38	15.62
Best capital growth	−9.26	87.11	24.39
Worst capital growth	−10.24	85.52	10.23

signals. Consequently, these signals cannot be fully utilized in the test data. Even when considering risk indicators, we are still unable to execute the strategy effectively.

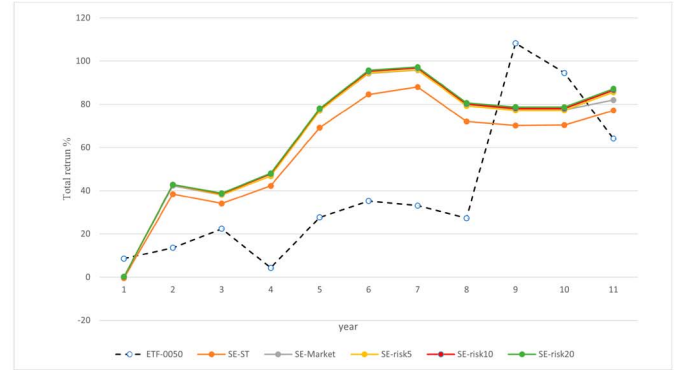
However, when the training period was extended to 240 days, the strategy’s performance improved significantly. The results indicate a substantial improvement in the capital growth rate when the test period is set to 120 days. However, the capital growth rate is less effective than expected when the test period is set to 60 or 240 days. We consider that when the test duration is too short, strategies, even if effective, must be updated every 60 days. However, frequent updates may reduce the overall effectiveness of the mechanism. If the number of test days is too long, the strategy may become ineffective over time due to infrequent updates. When the number of test days is 120, the strategies identified by SE are reliable, indicating that this setup optimally utilizes the training data and helps prevent underfitting.

TABLE V
ETF-0050 COMPARISON WITH OUR METHOD USING % SHOW RESULTS

Type	ETF-0050	SE-ST	SE-Market	SE-Risk5	SE-Risk10	SE-Risk20
Average capital growth	65.2	76.17	85.46	86.25	85.96	86.38
Average risk	24.87	15.41	16.32	16.37	16.37	16.33



(a)



(b)

Fig. 7. SE transaction calculation framework compare with 0050. (a) y-axis is represented in years. (b) y-axis is represented in half-year units.

Moreover, our results indicate that SE exhibits the lowest capital growth rate in the standard model. This result suggests that we infer that under the standard model, trading inevitably occurs in each test period. Compared with the market model, in which positions can be held until the final day of testing, frequent trading during the test period may lead to the accumulation of small losses, ultimately leading to poorer performance. Based on this result, we can further deduce that an overly aggressive trading strategy may still lead to unnecessary losses. In market mode and risk mode, the mechanism's design enables the strategy to respond to market conditions more efficiently, allowing it to capture trends and manage losses. As a result, the overall capital growth rate in these two modes surpasses that of the standard model.

Our comparison with the ETF-0050 is shown in Table V. As shown, our capital growth rate is generally higher with lower risk of excessive capital loss during the backtesting period. This methodology is effective in controlling losses during periods of poor market performance and identifies strategies to buy stocks at low points and sell them at high points to maximize capital growth. However, it is important to note that historical data cannot fully predict future market behavior, and economic fluctuations may introduce additional uncertainties. As seen in Fig. 7(b), our mechanism becomes less stable during periods of extreme market volatility. Nonetheless, excluding extreme years, our strategy remains competitive and consistently outperforms ETF-0050 in terms of capital growth.

In conclusion, our findings indicate that a 240-day training period with a 120-day test period provides the most effective balance for identifying stable trading signals. Although short-term strategies may involve higher risks, effective risk management significantly enhances performance. Investors should still consider the risk indicators when buying and selling stocks and should not trade too often. It is recommended that the stock be held for about half a year.

VI. CONCLUSION

In recent years, the surge in investment enthusiasm has led many Taiwanese investors to enter the market. However, the proportion of those who achieve profits in the stock market remains relatively low. Therefore, we propose a trading strategy based on the SE method. By leveraging SE's global search capabilities, we utilize historical data to identify effective trading strategies. Our experimental results demonstrate that our strategy successfully enhances capital growth. Therefore, this article proposes a framework for identifying stock signals based on SE. We aim to leverage SE's outstanding global search capabilities to develop effective trading strategies. Our methodology enables investors to significantly enhance their capital growth rate. This methodology is compared with ETF-0050 over a defined time horizon, within a framework designed to achieve a higher capital growth rate while minimizing risk during the investment period. In addition, we select stocks based on industry categories, and our methodology does not focus on a single stock within the Taiwanese market. This approach simulates the stock selection process of a typical investor in the market. Additionally, we select stocks based on industry categories rather than focusing on a single stock in the Taiwanese market. This approach simulates the stock selection process of a typical investor.

REFERENCES

- [1] H. Markowitz, "Portfolio selection," *J. Finance*, vol. 7, no. 1, pp. 77–91, 1952. Accessed: May 16, 2025. [Online]. Available: <http://www.jstor.org/stable/2975974>
- [2] E. F. Fama and K. R. French, "The cross-section of expected stock returns," *J. Finance*, vol. 47, no. 2, pp. 427–465, 1992. Accessed: May 16, 2025. [Online]. Available: <http://www.jstor.org/stable/2329112>
- [3] S. Siarni-Namini, N. Tavakoli, and A. S. Namin, "A comparison of Arima and LSTM in forecasting time series," in *Proc. 17th IEEE Int. Conf. Mach. Learn. IEEE Int. Symp. Spread Spectr.*

- Tech. Appl. (ICMLA)*, Piscataway, NJ, USA: IEEE Press, 2018, pp. 1394–1401.
- [4] J. Shah, D. Vaidya, and M. Shah, “A comprehensive review on multiple hybrid deep learning approaches for stock prediction,” *Intell. Syst. with Appl.*, vol. 16, 2022, Art. no. 200111.
 - [5] J. M.-T. Wu, Z. Li, N. Herencsar, B. Vo, and J. C.-W. Lin, “A graph-based CNN-LSTM stock price prediction algorithm with leading indicators,” *Multimedia Syst.*, vol. 29, no. 3, pp. 1751–1770, 2023.
 - [6] S. Oikonomopoulos, K. Tzafilkou, D. Karapiperis, and V. Verykios, “Cryptocurrency price prediction using social media sentiment analysis,” in *Proc. 13th Int. Conf. Inf., Intell., Syst. Appl. (IISA)*, Piscataway, NJ, USA: IEEE Press, 2022, pp. 1–8.
 - [7] S. M. Sarmiento and N. Horta, “Enhancing a pairs trading strategy with the application of machine learning,” *Expert Syst. Appl.*, vol. 158, 2020, Art. no. 113490.
 - [8] Vaswani, A “Attention is all you need,” *Advances in Neural Inf. Process. Syst.*, 2017.
 - [9] D. J. B. Sagi, F. J. Soltero, J. I. Hidalgo, P. Fernández, and F. Fernandez, “A technique for the optimization of the parameters of technical indicators with multi-objective evolutionary algorithms,” in *Proc. IEEE Congr. Evol. Comput.*, Piscataway, NJ, USA: IEEE Press, 2012, pp. 1–8.
 - [10] P. Sonkiya, V. Bajpai, and A. Bansal, “Stock price prediction using BERT and GAN,” 2021, *arXiv:2107.09055*.
 - [11] X. Weng, X. Lin, and S. Zhao, “Stock price prediction based on LSTM and BERT,” in *Proc. Int. Conf. Mach. Learn. Cybern. (ICMLC)*, Piscataway, NJ, USA: IEEE Press, 2022, pp. 12–17.
 - [12] A. Lopez-Lira and Y. Tang, “Can ChatGPT forecast stock price movements? return predictability and large language models,” 2023, *arXiv:2304.07619*.
 - [13] J. B. Chakole, M. S. Kolhe, G. D. Mahapurush, A. Yadav, and M. P. Kurhekar, “A q-learning agent for automated trading in equity stock markets,” *Expert Syst. Appl.*, vol. 163, 2021, Art. no. 113761.
 - [14] J. Kaplan et al. et al., “Scaling laws for neural language models,” 2020, *arXiv:2001.08361*.
 - [15] M. Abdel-Basset, L. Abdel-Fatah, and A. K. Sangaiah, “Metaheuristic algorithms: A comprehensive review,” in *Proc. Computat. Intell. Multimedia Big Data Cloud Eng. Appl.*, 2018, pp. 185–231.
 - [16] C.-W. Tsai, “Search economics: A solution space and computing resource aware search method,” in *Proc. IEEE Int. Conf. Syst., Man, Cybern.*, Piscataway, NJ, USA: IEEE Press, 2015, pp. 2555–2560.
 - [17] H.-H. Cho, M.-Y. Tsai, J.-Y. Zeng, C.-M. Yu, H.-C. Chao, and I. You, “LDoS attacks detection for ICPS NB-IoTs environment via SE-based CNN,” *IEEE Syst. J.*, vol. 17, no. 4, pp. 5280–5291, Dec. 2023.
 - [18] F. Yang, P. Wang, Y. Zhang, L. Zheng, and J. Lu, “Survey of swarm intelligence optimization algorithms,” in *Proc. IEEE Int. Conf. Unmanned Syst. (ICUS)*, Piscataway, NJ, USA: IEEE Press, 2017, pp. 544–549.
 - [19] J. Kennedy and R. Eberhart, “Particle swarm optimization,” in *Proc. Int. Conf. Neural Netw., (ICNN)*, Piscataway, NJ, USA: IEEE Press, 1995, pp. 1942–1948.
 - [20] Y.-H. Chou, S.-Y. Kuo, and Y.-C. Jiang, “A novel portfolio optimization model based on trend ratio and evolutionary computation,” *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 3, no. 4, pp. 337–350, Aug. 2019.
 - [21] Y.-H. Chou, Y.-C. Jiang, Y.-R. Hsu, S.-Y. Kuo, and S.-Y. Kuo, “A weighted portfolio optimization model based on the trend ratio, emotion index, and ANGQTS,” *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 6, no. 4, pp. 867–882, Aug. 2021.
 - [22] S.-Y. Kuo and Y.-H. Chou, “Building intelligent moving average-based stock trading system using metaheuristic algorithms,” *IEEE Access*, vol. 9, pp. 140383–140396, 2021.
 - [23] Y. Chen et al. et al., “A dynamic stock trading system using GNQTS and RSI in the US stock market,” in *Proc. IEEE Int. Conf. Syst., Man, Cybern. (SMC)*, 2021, Piscataway, NJ, USA: IEEE Press, pp. 456–461.
 - [24] F. D. Atiah and M. Helbig, “Effects of decision models on dynamic multi-objective optimization algorithms for financial markets,” in *Proc. IEEE Congr. Evol. Comput. (CEC)*, Piscataway, NJ, USA: IEEE Press, 2019, pp. 762–770.
 - [25] C.-H. Chen, Y.-H. Chen, J. C.-W. Lin, and M.-E. Wu, “An effective approach for obtaining a group trading strategy portfolio using grouping genetic algorithm,” *IEEE Access*, vol. 7, pp. 7313–7325, 2019.
 - [26] S.-Y. Kuo, C. Kuo, and Y.-H. Chou, “Dynamic stock trading system based on quantum-inspired tabu search algorithm,” in *Proc. IEEE Congr. Evol. Comput.*, Piscataway, NJ, USA: IEEE Press, 2013, pp. 1029–1036.
 - [27] R. de Almeida, G. Reynoso-Meza, and M. T. A. Steiner, “Multi-objective optimization approach to stock market technical indicators,” in *Proc.*

IEEE Congr. Evol. Comput. (CEC), Piscataway, NJ, USA: IEEE Press, 2016, pp. 3670–3677.

- [28] M. B. Fayek, H. M. El-Boghdadi, and S. M. Omran, “Multi-objective optimization of technical stock market indicators using gas,” *Int. J. Comput. Appl.*, vol. 68, no. 20, pp. 41–48, 2013.



Min-Yan Tsai received the B.S. and M.S. degrees from the Department of Computer Science and Information Engineering, National Ilan University, Yilan, Taiwan, in 2019 and 2022, respectively. He is currently working toward the Ph.D. degree with the Department of Information Management and Finance, National Yang Ming Chiao Tung University, Hsinchu, Taiwan.

His research interests include information security and artificial intelligence and evolutionary computing.



Jiang-Yi Zeng received the B.S. and M.S. degrees from the Department of Computer Science and Information Engineering, National Ilan University, Yilan, Taiwan, in 2021 and 2023, respectively. He is currently working toward the Ph.D. degree with the Department of Computer Science and Information Engineering, National Cheng Kung University, Tainan, Taiwan, since 2023.

His research interests include network security and artificial intelligence, artificial intelligence security, and evolutionary computing.



Chia-Mu Yu (Senior Member, IEEE) is currently an Associate Professor in the Department of Electrical Engineering at National Yang Ming Chiao Tung University (NYCU). He also holds joint associate professorships in the Department of Information Management and Finance, College of Artificial Intelligence, and the Industry Academia Innovation School at NYCU. His expertise lies in AI safety, data privacy, and network security. He has received numerous accolades, including the K. T. Li Pan-Shih Award, the National Science Council Young

Scholar Cultivation Program (Columbus Program), the NYCU Young Chair Professorship, the K. T. Li Young Research Award, the Pan Wen-Yuan Research Grant, and the National Science Council Outstanding Young Scholar Program.

He is currently a Senior Member of the IEEE and serves as an Senior Area Editor for IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY, and an Associate Editor for IEEE INTERNET OF THINGS JOURNAL AND IEEE CONSUMER ELECTRONICS MAGAZINE.



Fan-Hsun Tseng (Senior Member, IEEE) received the Ph.D. degree in computer science and information engineering from the National Central University, Taoyuan, Taiwan, in 2016.

In 2021, he joined the faculty of the Department of Computer Science and Information Engineering, National Cheng Kung University, Tainan, Taiwan, where he is currently an Associate Professor since 2024. His research interests include federated learning, reconfigurable intelligent surface, and intelligent reflecting surface.

Dr. Tseng received the Young Scholar Fellowship from the Ministry of Science and Technology in Taiwan in 2018, the Distinguished Young Scholar Award from the Computer Society of the R.O.C in 2020, the 2021 Best Young Professional Member Award from IEEE Tainan Section, and the 2021 K. T. Li Young Researcher Award from the Institute of Information & Computing Machinery. He is the Secretary General and the Chair of the Computer Society, Chapter of the IEEE Tainan Section. He has served as Editor-in-Chief of *International Journal of Agile and Extreme Software Development*, Associate Editor-in-Chief of *Journal of Computers*, Associate Editor of IEEE ACCESS, *Human-centric Computing and Information Sciences*, *Journal of Internet Technology*, *Journal of the Chinese Institute of Engineers*, *Wireless Networks*, and *IET Networks*.